4.4 Practical Investigation — Your Own Observation

What this workbook covers 本册讲什么

Spec 4.4: designing and conducting your own observational practical — design decisions (hypothesis, behavioural categories, coding sheets, sampling), the pilot study, controls and ethics, presenting quantitative data, running a chi-squared test on your results, thematic analysis of the qualitative data, and evaluating and improving your investigation.

How to use this booklet 怎么用

1. **Fill it in as you go** — check the answer key (last page) after each page.
2. **The bus worked example (p.3) and the chi-squared steps (p.6) matter most.** The exam asks you to describe YOUR procedure and YOUR data, so the example is your template.

Content map 内容地图

Page	Topic	What you must be able to do afterwards
p.2	The brief 要求	State what the practical must include (both data types, learned behaviour)
p.3	Worked example + hypothesis 公交车例	Retell the bus study; justify a non-directional hypothesis
p.4	Planning + coding sheet 规划	List the design decisions; draw a coding sheet; define behavioural categories
p.5	Pilot, controls, ethics + data 试点与数据	Say what a pilot is for; explain inter-rater reliability and public-place ethics; present the contingency table (bar graph + frequency)
p.6	Chi-squared 卡方	Compute expected values, χ^2, df, and compare with the critical value
p.7	Thematic analysis 主题分析	Name the five phases (Braun and Clarke) with bus examples
p.8	Conclusions + evaluation 结论评价	Write conclusions from both data types; evaluate on four dimensions
p.9	Improvements + checkpoint 改进真题	Answer the six improvement questions; textbook questions 4+2+2
p.10	Answer key + brain dump 答案	Mark honestly, then rewrite wrong answers in full

Revision tracker 复习打卡

☐ p.2–3 brief + example ☐ p.4–5 planning + data ☐ p.6 chi-squared ☐ p.7 themes ☐ p.8–9 evaluation + exam

The whole chapter in one sentence 一句话总结

You design a naturalistic observation of **learned behaviour**, tally pre-defined behavioural categories on a coding sheet for numbers, note richer detail for words, test the numbers with **chi-squared** (expected = row × column ÷ overall), sort the words into themes with **Braun and Clarke's five phases**, then defend your choices and fix your flaws.

The Brief: What You Must Do 要求

Two routes 两条路

Either conduct _____ observations, one for qualitative and one for quantitative data （two）, or conduct _____ observation that gathers both types at once （one）. The behaviour observed must be _____ （learned）, for example prosocial/antisocial behaviour, gender type behaviour, or modelling others.

What you must be able to produce 产出清单

Stage	Requirement
Design decisions	Behavioural _____, coding sheets and tallying, control, hypothesis construction, ethics and observer bias/effects （categories）
Data	Collect, present and comment on data: descriptive _____ and graphical representation （statistics）
Quantitative analysis	Analyse with a _____ test and explain the significance of the result （chi-squared）
Qualitative analysis	Conduct a _____ analysis on the qualitative data and describe findings （thematic）
Write-up	Write the _____, results and discussion sections of a report （procedure）

中文理解：这门实操考的是「设计决策链」

考试不考你真去坐了多少趟公交，考你能不能把每一步为什么这么选讲清楚：为什么选自然观察、为什么假设不定向、为什么用卡方。每个选择前面有理由，后面有代价，链条完整就是满分答案的骨架。

Your turn: choose your own behaviour 写你自己的

Your observation must target _____ behaviour （learned）. Possible targets: prosocial or antisocial behaviour, gender type behaviour, _____ others （modelling）.

The behaviour I will observe: _____

The learning theory that explains how it is learned (e.g. Bandura's social learning theory, learned by observing role models): _____

Two data types I will collect: quantitative = _____; qualitative = _____ （tallies on a coding sheet; written notes of what people said）

Worked Example + Hypothesis Decision

The worked example 公交车观察例

Observing gender differences in prosocial/polite behaviour on public _____ (transport). Prosocial behaviour is defined as any behaviour that is intended to _____ another (benefit). Data collected on an hour's bus journey between _____ on a Thursday (3.00 and 4.00 p.m.); in all ____ passengers observed, ____ women and ____ men (51; 24; 27).

Background literature 背景文献（决定假设方向的依据）

Source	Claim
Social learning theory	Gender-appropriate behaviour learned from role models (parents, teachers, siblings, peers, media) and reinforced through _____ and punishments (rewards); stereotypes such as women being kind and helpful may be modelled and rewarded more in women
Eagly (_____) (2009)	Women and men both engage in prosocial behaviour _____ (equally), but women show more _____ and relational prosocial behaviours, men more _____ ones (to gain status or show strength) (communal; agentic)
Leslie, Snyder and Glomb (2013)	Workplace _____ to charity: women donated more than men (donations)

The hypothesis decision 假设决策

Because the background research is _____ (mixed), a _____ (two-tailed) hypothesis was put forward (non-directional): there will be a _____ in the number of women and men who display prosocial behaviour by thanking the bus driver before alighting (difference). Rules of thumb: unsure of the effect → _____; previous results point one way → _____; either way the hypothesis must be clearly _____ (non-directional; directional; testable).

> **中文理解：假设方向不是拍脑袋**
> 文献吵得有来有回（Eagly 说两性都做但类型不同，Snyder 说女性捐得多），没有一边倒的证据，就老实选不定向。考试问「为什么你的假设是不定向的」，答案就是这句：背景文献 mixed。如果文献一致指向一个方向，才选定向。

Quick check 快问

In Eagly's terms: quietly giving up a seat to help someone = _____ prosocial behaviour; helping loudly so others see your strength = _____ (communal; agentic)

Write your own non-directional hypothesis for one behavioural category on a bus:

Type of observation 观察类型

Observing in a natural setting (a real bus) means a _____ observation （naturalistic）. Sampling technique: participants board by chance, so _____ sampling （opportunity）. Design decisions also needed: which bus or train _____, the day of the week, the _____ of day and the duration of the journey （route; time） — all influence how many passengers and what type.

Behavioural categories 行为类目

Measure polite behaviour as whether passengers _____ the driver before alighting （thank）. Other possible categories: assisting other passengers, giving up a _____, making way for others on a crowded bus （seat）.

Coding sheet（Table 25.1）计数表

Behaviour	Male	Female
Thanking the bus driver	_____ (I (1))	_____ (III (3))
Giving up a seat on the bus	_____ (II (2))	_____ (—)
Helping a passenger with a pushchair	_____ (I (1))	_____ (II (2))

Tallying counts give _____ data, testable with an inferential test （quantitative）. For _____ data, write down what passengers actually said, the driver's gender, ages, eye contact （qualitative） — analysed later with thematic analysis.

> **中文理解：coding sheet 是双轨数据的接口**
> 同一趟观察两条轨道：计数栏记频率（定量），便签栏记细节（定性）。类目必须操作化定义清楚（什么叫「感谢司机」：下车前对司机说谢谢，方向对、时机对就算），否则两个观察员记的不是同一个东西，信度就崩了。

Operationalise it 下操作定义

Write an operational definition of 'thanking the driver' (state exactly what counts, said by whom, and when):

Which type-words apply to the bus observation? Naturalistic, because the setting is _____; likely covert, because passengers are _____; likely non-participant, because the observer _____ （natural; unaware; does not join in）.

Pilot Study, Controls, Ethics + Presenting the Quantitative Data 定量数据

Pilot study 试点研究

Run before the main observation to address potential problems and test your observational _____ （criteria）. Questions it answers: where is the best _____ to observe (sit near the front for driver interactions) （position）; how long the journey needs to be for enough data; whether you can observe everyone at busy times. **Key point:** a pilot is not used to test whether the investigation is _____ or _____ （valid; reliable） — it only checks apparatus and timings work correctly. It is also a chance to ask participants whether they felt uncomfortable or suspected anything.

Controls 控制

Recruiting a second observer helps, e.g. one watches _____ passengers, one watches _____ （male; female）. You must then consider _____ reliability: the degree of agreement between two or more observers （inter-rater）, ensuring consistency and controlling for _____ bias （observer）. The observation is likely _____ (participants unaware) （covert）; check whether passengers notice your note-taking and become suspicious — that would pose a _____ variable （confounding）.

Ethical considerations 伦理

Observations in _____ places are considered acceptable because you would expect to be observed in day-to-day interactions there （public）. You collect no names or personal data, only gender and observed behaviour, so no ethical guidelines are _____ （contravened）. You might gain consent after the observation is complete, though this may not be _____ （practical）.

中文理解：pilot 的边界最容易考
一句话卡死：pilot 查流程行不行（器材、时长、站位），查信度效度是主实验的事。公共场合观察伦理过关的理由也记两条：本来就默认会被看见；只记性别和行为，不碰个人信息。

Contingency table 列联表（Table 25.2）

	Men	Women	Total
Thanked the bus driver	_____ (cell A) （11）	_____ (cell B) （18）	_____ （29）
Did not thank the driver	_____ (cell C) （16）	_____ (cell D) （6）	_____ （22）
Total	_____ （27）	_____ （24）	_____ （51）

This is a _____ contingency table, suitable for _____ data （two-by-two; nominal）.

Descriptive statistics and graphs 描述统计与图

Best descriptive statistic for this data: _____ （frequency）. Best graph: a _____ graph, because nominal level data （bar）. Remember to label the _____ and axes fully and provide a key where necessary （title）.

中文理解：数据类型决定图表
名义数据只有频次数，均值中位数都算不出来，所以描述统计用 frequency，画图用 bar graph。散点图要两列连续数据，折线图要时间序列，都对不上。这条选择题课本 checkpoint 原题考过。

Running the Chi-Squared Test 卡方实操

Why chi-squared 为什么选卡方

The study predicts a _____, uses unrelated data / _____ groups, and the data is _____ level （difference; independent; nominal）. The full nine-step method is in 4.2; here is the compressed three-step version used in the practical.

Step 1: Expected values 期望值

Formula: E = row total × column total ÷ _____ total （overall）

Cell	Calculation	Result
A	_____ （29 × 27 / 51）	_____ （15.35）
B	_____ （29 × 24 / 51）	_____ （13.65）
C	_____ （22 × 27 / 51）	_____ （11.65）
D	_____ （22 × 24 / 51）	_____ （10.35）

Step 2: (O – E)² ÷ E per cell, then add 加总

Cell	Calculation	Result
A	(_____ – 15.35)² ÷ 15.35 （11）	_____ （1.23）
B	(_____ – 13.65)² ÷ 13.65 （18）	_____ （1.39）
C	(_____ – 11.65)² ÷ 11.65 （16）	_____ （1.62）
D	(_____ – 10.35)² ÷ 10.35 （6）	_____ （1.83）
Total	χ^2 = sum of the four cells	_____ （6.07）

Step 3: df, critical value, decision 自由度与判定

df = (_____ – 1) × (_____ – 1) = _____ （rows; columns; 1）. For a non-directional/two-tailed hypothesis, df = 1, $p < 0.05$, the critical value is _____ （3.84）. Observed _____ critical value → significant, reject the _____ hypothesis （6.07 ≥ 3.84; null）. Less than a five per cent probability the difference is due to _____ （chance）. Direction found by examining the cells: _____ were more likely to thank the bus driver （females）.

Chi-squared has a _____ column, not an 'n' column, because nominal data gives no individual scores （df）.

中文理解：三步就能交卷

算期望（行乘列除总数）→ 算卡方（差方除期望再相加）→ 查表判定（自由度行减一乘列减一，观测值大于等于临界值就显著）。方向看格子：18 比 6，女的更爱说谢谢。4.2 讲完整九步法，这里练三步压缩版，考试两边都要会。

The five phases 五阶段

Phase	What happens	Bus example
1. Familiarisation 熟悉数据	Read through the notes, start noticing _____ (themes)	Males say 'Thanks' and make small talk ('What time does your shift finish?'); females say thank you plus a _____ greeting and make eye contact (departure)
2. Generating initial codes 生成初始编码	A label or code given to specific categories identified	'thank you, with _____', 'thank you, with no familiarity', 'departure greeting', 'small talk' (familiarity)
3. Searching for themes 寻找主题	Seek themes on the basis of initial codes for some meaning	'_____ familiarity' — passengers who make small talk treat the driver as familiar (implied)
4. Reviewing themes 回顾主题	Try out the categories, e.g. collect _____ set of data to see if future observations fit (another)	Does a second bus journey produce the same patterns?
5. Defining and naming themes 定义命名	Themes clearly defined; a _____ is produced telling a story about the emerging themes (report)	Gender differences in how politeness is expressed to drivers

Qualitative vs quantitative data 两种数据的性格

Qualitative data cannot be summarised in a chart or graph; it provides much _____ detail on the complexities of human interaction (richer). But it is harder to select patterns and draw _____ conclusions (firm), and it is more likely to be _____, reflecting the researcher's personal viewpoints and background (subjective).

中文理解：五阶段记五个字（读编找验定）

先通读材料有感觉，再给片段贴标签，标签归类成主题，拿新数据验主题站不站得住，最后给主题命名写成报告。考点常考两处：Braun and Clarke 配 2006；第四阶段是回头验证，顺序不能乱。

Conclusions 结论

_____ data: the chi-squared result is significant, so the null hypothesis is rejected and the experimental hypothesis accepted （Quantitative）. Qualitative data: draw conclusions from the _____ in the observation, supporting your interpretation with specific _____ or observations （themes; quotes） (e.g. more small talk with male passengers, backed by actual quotes).

Evaluation: four dimensions 四个维度

Dimension	Points
Validity 效度	Naturalistic setting in real life suggests a degree of validity. 'Thank you' is a relatively _____ measure (no judgement involved) （objective）; but qualitative notes like 'small talk' involve more _____ （interpretation）. Improvement: _____ validation — get feedback from participants (e.g. interview regular bus users) to check your interpretation （respondent）
Reliability 信度	Can another observer repeat the study and get similar findings? _____ reliability: agreement between two or more observers; lack of consensus means the behavioural categories may need to be _____ （inter-rater; revised）. _____–retest: repeat the observation on the same route later (a day, week or month) and compare （test）
Generalisability 概化	Time of day affects the _____ achieved （sample）; ages need consideration — an equal distribution minimises bias. More retired passengers may raise polite interactions (less time-pressured, more traditional values of respect)
Credibility 可信度	Depends on scientific principles: _____, measurable phenomena （replicability）. Weigh overall strengths (naturalistic = valid measurement) against weaknesses (fixed time of day limits generalisation; repeated observations may be needed)

Triangulation 三角互证

_____: using more than one approach to investigate a research question （Triangulation）. A single method suffers from that method's own limits; checking results by other methods gives increased _____ in the findings if similar results emerge （confidence）. Example: a _____ with bus passengers on politeness, to see if their views match the behaviour observed （questionnaire）.

中文理解：四维评价一句话版

效度看情境真不真、类目客观不客观；信度看换个观察员换个日子还能不能复制；概化看时段和年龄偏不偏；可信度看符不符合科学原则。respondent validation 和 triangulation 是两个加分招：一个问当事人认不认，一个换方法对不对得上。

Six improvement questions 改进六问

1. Was the _____ representative? How could it be made more representative? (sample)
2. Was the _____ natural? How could it be more natural? (situation)
3. Any issues with lack of _____? What controls could be used? (control)
4. Were the behavioural categories measuring what was _____? How could they be operationalised more effectively? (intended)
5. Was any data collection or analysis _____? How could it be more objective? (subjective)
6. Any unforeseen _____ issues? How could ethics be improved? (ethical)

Checkpoint（课本）

1. Put the chi-squared instructions in the correct order: a) subtract the expected value from the observed value, then divide by the expected value · b) compare the calculated value with the critical/table value · c) calculate expected values (row total × column total / overall total) · d) total the scores for each cell to give the chi-squared value.

First _____ then _____ then _____ finally _____

2. Best descriptive statistic for the quantitative data: a) mean b) median c) frequency → _____

3. Best graphical presentation: a) scattergraph b) bar graph c) line graph → _____

Exam practice（课本）

1. (a) Describe the procedure of your learning theories and development practical investigation. (4) (b) Explain one strength of your practical investigation in terms of validity. (2) (c) Explain one improvement you could have made in terms of reliability. (2)

Answer key 答案速查

p.2 two · one · learned · categories · statistics · chi-squared · thematic · procedure · learned/modelling · tallies on a coding sheet / written notes.
p.3 transport · benefit · 3.00–4.00 p.m. · 51/24/27 · rewards · 2009 · equally · communal · agentic · donations · mixed · non-directional · difference · non-directional; directional; testable · communal; agentic (quick check).
p.4 naturalistic · opportunity · route · time · thank · seat · I/III · II/— · I/II · quantitative · qualitative · operational definition = own wording · natural · unaware · does not join in.
p.5 criteria · position · valid · reliable · male · female · inter-rater · observer · covert · confounding · public · contravened · practical · 11/18/29 · 16/6/22 · 27/24/51 · two-by-two · nominal · frequency · bar · title.
p.6 difference · independent · nominal · overall · 29×27/51→15.35 · 29×24/51→13.65 · 22×27/51→11.65 · 22×24/51→10.35 · 11/18/16/6 · 1.23/1.39/1.62/1.83 · 6.07 · rows · columns · 1 · 3.84 · 6.07 ≥ 3.84 · null · chance · females · df.
p.7 themes · departure · familiarity · implied · another · report · richer · firm · subjective.
p.8 Quantitative · themes · quotes · objective · interpretation · respondent · inter-rater · revised · test · sample · replicability · Triangulation · confidence · questionnaire.
p.9 sample · situation · control · intended · subjective · ethical · order: c, a, d, b · 2c · 3b.

Model answer highlights 高分要点（4+2+2 分题）

Q1(a) 描述自己的 procedure（4 分骨架）

Describe what you actually did in order: naturalistic observation of passengers on a bus (state route, day, time, duration); behavioural category = thanking the driver before alighting; tallied males and females on a coding sheet (quantitative) while noting what passengers said (qualitative); opportunity sampling; 51 passengers observed. Four distinct steps, each with a detail.

Q1(b) 效度优点 + Q1(c) 信度改进（2+2 分骨架）

(b) Validity strength: the observation was naturalistic, carried out in a real bus where the behaviour normally occurs, so thanking behaviour was genuine and spontaneous. (c) Reliability improvement: recruit a second observer and check inter-rater reliability (degree of agreement between observers); disagreement would show the behavioural categories need revising. State the concept + apply it to YOUR study.

Brain dump 空白重写

Rewrite, from memory and in full, any answer you got wrong or left blank above: